

SHIHAB BIN KADER

Artificial Intelligence | Computer Vision | Software Development

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PROFESSIONAL PROFILE

Computer Science and Engineering graduate with hands-on experience in machine learning, computer vision, deep learning, explainable AI, and software development. Built academic and open-source projects using Python, PyTorch, TensorFlow/Keras, OpenCV, Streamlit, Java, SQL, and Git/GitHub. Comfortable with data preparation, model evaluation, debugging, documentation, and collaborative project work.

EDUCATION

BSc in Computer Science and Engineering, East Delta University

Sep 2020 - Jan 2025

Chattogram, Bangladesh | CGPA: 3.13/4.00 | 162 credits

Joint thesis: Mango Ripeness Detection Using Deep Learning with XAI (Grad-CAM)

Relevant coursework: Artificial Intelligence, Machine Learning, Pattern Recognition, Digital Image Processing, Probability and Statistics, Algorithms, Database Management Systems, and Software Engineering.

ACADEMIC & RESEARCH EXPERIENCE

Joint Undergraduate Thesis Researcher, East Delta University

2024 - 2025

- Co-conducted a six-stage mango-ripeness study using a balanced 6,000-image dataset, preprocessing and augmentation, and an 80/10/10 train/validation/test design.
- Compared CNN and transfer-learning architectures using accuracy, precision, recall, F1-score, confusion matrices, ROC/AUC, class-wise error analysis, and Grad-CAM explanation maps.

PUBLICATION

Abdullah, M. S., Faiyaz, G. M. F., Forhad, O. N., Kader, S. B., & Labib, S. M. A. (2025). "Hybrid Xception-LSTM Model for Explainable Mango Ripeness Classification Using Grad-CAM." IEEE 4th RAAICON, Dhaka, Bangladesh. [Fourth author]

SELECTED OPEN-SOURCE & TECHNICAL PROJECTS

ModelDNA - Explainable AI Toolkit | Python, PyTorch, Streamlit, OpenCV, Grad-CAM

- Developed an interactive toolkit for class-activation heatmaps, confidence analysis, and comparison of visual attention patterns.

Sign Sense - Real-Time ASL Detector | Python, TensorFlow/Keras, OpenCV

- Developed a 29-class webcam recognition application and examined confusion among visually similar gesture classes.

Drowsiness Detection AI | Python, TensorFlow/Keras, OpenCV

- Implemented real-time eye-state classification with attention to streaming images, latency, confidence, and changing visual conditions.

Image Caption Generator | Python, InceptionV3, LSTM

- Implemented a visual-language pipeline combining pretrained image features with sequence generation and qualitative error analysis.

Multimodal AI Application | Python, Streamlit, Hugging Face Transformers

- Integrated image captioning, PDF question answering, and chatbot workflows in one interface for heterogeneous inputs.

Fun with Flags - Android Quiz App | Java, Android Studio, MySQL

- Built an interactive mobile quiz with score tracking, result reporting, and database-backed data handling.

TECHNICAL SKILLS

Programming and data: Python, Pandas, NumPy, Java, C, C++, SQL, JavaScript/TypeScript

Machine learning and vision: PyTorch, TensorFlow/Keras, OpenCV, CNNs, LSTMs, transfer learning, Grad-CAM, feature extraction, confidence analysis

Development: Streamlit, FastAPI, React, MySQL/PostgreSQL, Docker, Android Studio, Git/GitHub

Research and work practices: dataset curation, augmentation, model comparison, debugging, literature review, documentation, and reproducible reporting

PROFESSIONAL INTERESTS

Computer Vision | Explainable AI | Multimodal Learning | AI Agents | Applied Machine Learning | Software Development

VOLUNTEERING

Community Volunteer, Chattogram | Present - Supported educational and social programs for children and helped organize community activities promoting child welfare and development.

HONOURS & AWARDS

Dean's List Recognition, East Delta University | EDU Academic Scholarship for academic performance

ENGLISH PROFICIENCY

IELTS Academic: Overall 7.5 (CEFR C1) | Listening 8.0 | Reading 8.0 | Writing 6.5 | Speaking 7.0 | January 2026

LANGUAGES

Bengali: Native | English: Advanced (IELTS 7.5, CEFR C1)